Natural-Language Controls for Short-Video Recommenders

An offline study of controllability

Vibhor Malik

Supervisor review draft: 14 October 2026
COMP 9500, M.Sc. Research Project

British Columbia Institute of Technology
Supervisor: Borna Noureddin

AI assistance was used for implementation, analysis and drafting.

Current research position

Experiment 1: category enforcement under a fixed policy.
No LLM ranking advantage on canonical templates.

Exposed practice: better Llama control success,
with lower NDCG@10 than item feedback.

Formal unseen E2: replacement bank and eligible
independent annotation remain pending.

No unseen language-generalization, recommendation-accuracy gain or human-experience benefit is established.

Research question

Can one natural-language interaction satisfy a richer intended change than one item reaction?

- A compound request can specify several categories, strength and exceptions.
- The LLM must also outperform a rule parser to establish its own contribution.
- Offline compliance and historical prediction measure different outcomes.

The claim that language *feels more natural* requires human evidence beyond this study.

Prior work supports a control question

Source	Relevant evidence
Mozilla, 2022	Dislike prevented 12% of unwanted recommendations and Not interested 11%, relative to no feedback.
Ramos et al., 2024	Positive profile edits raised target Coverage@10 without further fine-tuning. Amazon-MT RMSE: 0.941 versus MF 0.925, where lower is better.
Sanner et al., 2023	Language preferences were competitive in a near cold-start movie setting.
Chen and Pu, 2012	Critiquing systems already studied richer explicit preference feedback.

Historical results in different settings. Full references and source links: backup slide and speaker notes.

Translation and enforcement are separate

Stage	Role
Recommender	Frozen SASRec-style scores for the candidate catalogue.
Translator	Llama 3.1 8B or rules compile language into a validated command.
Enforcer	Shared deterministic reranking applies the command.
Explanation	Verified facts describe the displayed item and control.

A correct hard exclusion comes from the enforcement code. It does not demonstrate language understanding.

Data and information boundaries

- KuaiRec cohort: **865 users**, **2,593 candidates**, 31 opaque category identifiers.
- Positive event: watch ratio > 2, an engagement proxy.
- Global chronological windows: nominal 70% training, 15% request history, 15% held-out evaluation.
- Requests use training and request history only.
- Quality target: first eligible held-out positive item.

Synthetic requests and numeric category labels limit claims about real preferences and language.

Filtered cohort counts, not the upstream KuaiRec matrix size. Frozen candidate policy permits seen items.

Five matched configurations

ID	Mechanism
A	One simulated reaction changes one historical exemplar item.
B	A plus display-only explanations.
C	Free text compiles into a category control.
D	An editable preference sentence compiles into a category control.
E	Text, profile and explanations, with duplicate commands removed.

One boost and one mute request per user, under seeds 42, 43 and 44. The item-feedback proxy is an explicit design choice.

A is not a reconstruction of a commercial feedback algorithm. B must rank identically to A.

Parser results on synthetic language

Language material	Llama correct	Meaning
Templated inputs	120/120	Canonical task
Broader synthetic set	79/100	Below 85% target
Adversarial subset	2/10	Major weakness

- 79% exceeds the 70% floor. It misses the 85% target.
- The rule parser scores 81/100 on the same broader set.
- The 120 canonical wording patterns informed development.

The adversarial subset belongs to the 100 cases. These finite synthetic sets do not establish general language reliability.

Category enforcement works under the policy

Top-10 exposure	A	C, D and E
Boost target share, TCP	11.6%	79.2%
Muted target share, TCER	6.1%	0.0%

- Hard exclusion guarantees zero exposure when parsed correctly.
- Category-wide boosts affect more items than A's single-item adjustment.
- Rule and LLM paths produce identical feeds for this task.

Averages over 865 users and three seeds. This is enforcement evidence, not an LLM advantage.

Steering has a predictive quality cost

Boost requests	A	C
NDCG@10	0.03012	0.01637

45.7% relative decrease.

Paired mean change: -0.01375

95% user-bootstrap interval: $[-0.02086, -0.00655]$.

This post-hoc secondary contrast measures agreement with a historical next item. It does not measure user dissatisfaction after steering.

Post-hoc exploratory interval, outside the frozen primary BH family.

A weak backbone limits the quality claim

Uncontrolled ranking	HR@1
SASRec, seed 42	0.000%
SASRec, seed 43	0.347%
SASRec, seed 44	0.462%
Recency popularity	7.977%

Recency pools all users' request history. The model's weights use the older training window. This comparison is descriptive. The absolute 10 percentage point non-inferiority margin is uninformative at this baseline.

Post-hoc uncontrolled baselines. A pooled HR@1 interval includes zero. No accuracy gain is established.

Experiment 1 conclusion

1. Deterministic category control works: TCP 11.6% to 79.2%, TCER 6.1% to zero.
2. Llama parsed 120/120 templated inputs, 79/100 broader synthetic inputs and 2/10 adversarial cases.
3. All 10/10 explanation fact selections equal the template.
4. Boost NDCG@10 falls 45.7%. Non-inferiority is uninformative.
5. Backbone HR@1 is 0.0–0.46%, below recency popularity's 8.0%.

The study establishes bounded enforcement and its limitations. It does not establish an LLM advantage or useful quality preservation.

Experiment 2: richer intent per interaction

H1 Higher induced intent satisfaction than both the one-item proxy and the rule parser.

H2 More intended-direction rank movement than one item reaction, with one interaction each.

H3 Paired NDCG@10 and HR@1 changes against A and recency popularity, with intervals.

H3 is descriptive. It has no pass threshold and no non-inferiority claim.

Prespecified protocol dated 29 September 2026. H1 and H2 remain unassessed.

Four classes of intent

Class	Illustrative request	Top-10 requirement
Compound	More X and less Y	$n(X) \geq 2$, $n(Y) \leq 1$
Graded	A bit more X	$2 \leq n(X) \leq 4$
	Much less X / none of X	$n(X) \leq 1$ / $n(X) = 0$
Floor	Less X, but not zero	$1 \leq n(X) \leq 2$
Conditional	More X only if also Y	$n(X \cap Y) \geq 2$ and no increase in $X \setminus Y$ count

Success also requires a full list and positive intended movement. Bands are operational conventions.

X and Y denote opaque categories. Conditional means the same item carries both labels. These are illustrative wording forms, not reserved cases.

A matched interaction budget

- A gets one positive or negative reaction on one historical exemplar.
- Rule and LLM get the complete utterance once and share the same ranker.
- Assignments use only earlier history and a committed SHA-256 scheme.
- Channels share permitted pre-test information. Recency and SASRec use that information differently.

The budget matches interaction count. It does not equalize semantic content, user effort or every possible feedback algorithm.

Formal evaluation procedure

1. Preserve the original bank and September 29 commitment.
2. Agree a replacement unseen bank and annotation route.
3. Obtain independent labels, agreement and blinded adjudication.
4. Complete the full freeze and verify model provenance.
5. Execute once; report parser results before ranking.

The current 100 worked questions and AI answers are exposed practice. Formal E2 remains pending.

Development evidence and bank exposure

Development v1: rules 48/48; Llama 24/48.
Ten Llama operational errors were retained.

Development v2: rules 48/48; Llama 48/48.
Prompt and wording changed during development.

The original 100 questions are now exposed practice.
No unseen-test or formal H1/H2 result is claimed.

Archived September 29 development evidence is preserved. Equal development scores establish no LLM advantage.

Exposed practice execution on October 14

AI reference agreement: Llama 73/100; rules 18/100.
Llama validation rejections: 24/100, retained.

Induced control success: Llama 23.81%; rules 8.40%;
item feedback 3.92%.
NDCG@10: Llama 0.02461; item feedback 0.03128.

41,508 ranking rows; 865 users; three frozen seeds.

Exposed AI references, not unseen human gold. Fixed Llama 3.1 8B / Ollama 0.34.0; no tuning. Metric recount passed.

Prespecified analysis

- Unit: user. Average three seeds within each user.
- H1: four classes and pooled, against A and rules.
- H2: pooled intended movement against A.
- One family of **11** Wilcoxon comparisons, BH correction.
- User-bootstrap intervals from **5,000** resamples.
- Retain impossible and already-satisfied cases with explicit flags.

H1 needs both positive pooled contrasts with $q < 0.05$. H2 needs its positive pooled contrast with $q < 0.05$.

Signed-rank symmetry, tied/sparse differences and BH dependence assumptions remain limitations.

Current completion limits

Experiment 1: complete and frozen; 288 hashes match.

Practice diagnostic: parsing and ranking completed.
A separate metric recount found no discrepancies.

Formal E2: replacement bank and human evidence pending.
H1/H2 unassessed; formal H3 has not run.

Windows practice verified; full E1 replay and Colab unverified.

Formal study requires genuinely independent annotation, agreement and blinded adjudication.

Contribution and publication direction

Supported contribution

An inspectable pipeline separates translation from enforcement.

Frozen E1 establishes bounded enforcement and quality cost.

Practice records a control benefit alongside NDCG cost.

Work needed for a stronger paper

Approve a fresh bank and valid annotation route prospectively.

Complete the full freeze before one formal evaluation.

Human experience needs a separately approved extension.

The exposed practice result is descriptive; formal H1/H2/H3 remain outstanding.

Proposed 135-hour completion plan

Proposed period	Main checkpoint
5–23 Oct	Scope, sources, annotation and runtime. Readiness decision by 23 October.
26 Oct–8 Nov	Reserved evaluation 26 Oct–1 Nov if ready. Analysis 2–8 November.
9–27 Nov	Reproduction 9–15 Nov. Cold review 16–22 Nov. Complete paper draft by 27 Nov.
30 Nov–18 Dec	Supervisor revisions, repository handoff, rehearsal and final review.

11 remaining calendar weeks, **135 prospective hours**. Exact submission date requires confirmation.

Weekly check-in: Wednesday 10:00 am. Update by Tuesday 10:00 am. Meeting notes follow each meeting. No completed hours are claimed here.

Supervisor discussion

1. Decide the prospective amendment and approval record.

2. Agree fresh-bank provenance, access and annotators.

3. Resolve the original non-inferiority margin wording.

4. Confirm final date and repository/evidence access.

The main report is within 6-8 pages including references;
the extended report and audit evidence are separate.

Practice parsing/ranking is verified. No unseen bank, eligible human labels or formal E2 completion is claimed.

Backup: metric definitions

TCP@10 Target-category items divided by ten for boost requests.

TCER@10 Target-category items divided by ten for mute requests.

HR@1 One if the first item equals the held-out target, otherwise zero.

NDCG@10 For one relevant item at rank $r \leq 10$, $1/\log_2(r+1)$. Zero if absent.

Both exposure metrics have the same arithmetic and opposite desired directions. Historical quality does not directly measure preference satisfaction after intervention.

Backup: backbone and control settings

Backbone	Compact SASRec-style implementation
Hidden size / context	32 / 50 items
Attention	Two blocks, one head
Training	20 epochs, Adam, learning rate 0.001
Batch / dropout	64 / 0.2
Training seeds	42, 43, 44
Experiment 1 control	Normalized score +0.25, or hard mute
Experiment 2 strengths	0.10, 0.25, 0.50
Language backend	Llama 3.1 8B, temperature 0

The backbone is independently implemented. The study does not claim an exact SASRec paper reproduction.

Backup: why non-inferiority is uninformative

Pooled Experiment 1 quantity	Value
A HR@1 baseline	0.2505%
C minus A, mean	$+0.193$ pp
Paired 95% interval	$[-0.058, +0.501]$ pp
Simultaneous lower bound	-0.07707 pp
Absolute proposed threshold	-10 pp
Relative 10% loss threshold	-0.02505 pp

The absolute reading numerically passes while allowing the entire small baseline to disappear. The relative reading fails. Only eight of 865 users have nonzero paired HR@1 differences.

Both operations pooled, distinct from the boost-only NDCG contrast. pp means percentage points.

Backup: intended-direction movement

$$D(S) = \frac{\sum_{r=1}^{10} \mathbf{1}[i_r \in S] / \log_2(r+1)}{\sum_{r=1}^{10} 1 / \log_2(r+1)}$$

ΔD is the channel value minus its uncontrolled value.

Class	Signed movement M
Compound	$\min(\Delta D(X), -\Delta D(Y))$
Boost / reduction	$\Delta D(X)$ / $-\Delta D(X)$
Floor	$-\Delta D(X)$, zero if the floor is violated
Conditional	$\Delta D(X \cap Y) - \max(0, \Delta D(X \setminus Y))$

Induced satisfaction requires a full list, the class band and $M > 10^{-12}$. This operational measure is not Shannon information.

Backup: evidence and responsible authorship

Frozen E1: preserved data, checkpoints, prompts and outcomes.
All 288 manifest hashes still match.

Practice: exact inputs, AI-reference provenance, raw responses, full scores, ranking metrics and failures are saved.

Validated parser score: 73/100, with all 24 errors retained.
Control benefit and NDCG cost are reported together.

Human annotation and the fresh formal bank remain pending.

Personal reading, independent annotation and activity hours are recorded only when actually completed.

Backup: selected references

- Ricks and McCrosky (Mozilla), 2022. *Does this button work? Investigating YouTube's ineffective user controls.*
- Ramos et al., 2024. *Transparent and Scrutable Recommendations Using Natural Language User Profiles*. arXiv:2402.05810.
- Sanner et al., RecSys 2023. *Large Language Models are Competitive Near Cold-start Recommenders for Language- and Item-based Preferences*. arXiv:2307.14225.
- Chen and Pu, 2012. *Critiquing-based recommenders: survey and emerging trends*. UMUAI 22, 125–150.
- Kang and McAuley, ICDM 2018. *Self-Attentive Sequential Recommendation*.
- Gao et al., CIKM 2022. *KuaiRec: A Fully-observed Dataset and Insights for Evaluating Recommender Systems*.

Complete bibliographic records: paper/references.bib. Direct source links: speaker-notes.md.